UNIFORM BOUNDEDNESS PROPAGATION ALONG SYSTEMS OF PERFECT COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. One named auxiliary 'details omitted' (faithfulness of pr_{2,*} on supported modules) transcribed as-is; the main direction is complete in-span.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 2143 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Kollar-Kovacs pseudo-coherence. more-algebra-lemma-kollar-kovacs-pseudo-coherent (cross-chapter, named). Fibre-wise boundedness lifts to the complex.
- Split on regular schemes. lemma-split-complex-regular L26491 (carded this batch). Gives the direct sum decomposition for n large.
- Smooth regularity. varieties-lemma-smooth-regular (cross-chapter, named). X x Y is regular in the smooth case.

Source reference. Original lemma 0FZ4: sources/stacks-equiv/pinned-original-source.tex, source lines 1920–1920. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0EGV: focused-seeds/geometry/S9-eqv-bounded-fibres/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 056S: focused-seeds/geometry/S9-eqv-bounded-fibres/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let X, Y be finite type schemes over k. Let $K_0 \to K_1 \to K_2 \to \ldots$ be a system of objects of $D_{perf}(\mathcal{O}_{X\times Y})$ and $m \geq 0$ an integer such that*

(1) *$H^q(K_i)$ is nonzero only for $q \leq m$,*

Date: stacks-project:equiv.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *for every coherent $\mathcal{O}_X$-module $\mathcal{F}$ with* $\dim(Supp(\mathcal{F})) = 0$ *the object*

$$Rpr_{2,*}(pr_1^*\mathcal{F} \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times Y}} K_n)$$

has vanishing cohomology sheaves in degrees outside $[-m, m]\cup[-m-n, m-n]$ *and for* $n > 2m$ *the transition maps induce isomorphisms on cohomology sheaves in degrees in* $[-m, m]$.

Then K_n has vanishing cohomology sheaves in degrees outside $[-m, m] \cup [-m - n, m - n]$ and for $n > 2m$ the transition maps induce isomorphisms on cohomology sheaves in degrees in $[-m, m]$. Moreover, if X and Y are smooth over k, then for n large enough we find $K_n = K \oplus C_n$ in $D_{perf}(\mathcal{O}_{X\times Y})$ where K has cohomology only indegrees $[-m, m]$ and C_n only in degrees $[-m - n, m - n]$ and the transition maps define isomorphisms between various copies of K.

Proof. Let Z be the scheme theoretic support of an $\mathcal{F}$ as in (2). Then $Z \to \mathrm{Spec}(k)$ is finite, hence $Z\times Y \to Y$ is finite. It follows that for an object M of $D_{QCoh}(\mathcal{O}_{X\times Y})$ with cohomology sheaves supported on $Z\times Y$ we have $H^i(R\mathrm{pr}_{2,*}(M)) = \mathrm{pr}_{2,*}H^i(M)$ and the functor $\mathrm{pr}_{2,*}$ is faithful on quasi-coherent modules supported on $Z \times Y$; details omitted. Hence we see that the objects

$$\mathrm{pr}_1^*\mathcal{F} \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times Y}} K_n$$

in $D_{perf}(\mathcal{O}_{X\times Y})$ have vanishing cohomology sheaves outside $[-m, m]\cup[-m-n, m-n]$ and for $n > 2m$ the transition maps induce isomorphisms on cohomology sheaves in $[-m, m]$. Let $z \in X \times Y$ be a closed point mapping to the closed point $x \in X$. Then we know that

$$K_{n,z} \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times Y,z}} \mathcal{O}_{X\times Y,z}/\mathfrak{m}_x^t\mathcal{O}_{X\times Y,z}$$

has nonzero cohomology only in the intervals $[-m, m]\cup[-m-n, m-n]$. We conclude by More on Algebra, Lemma 0EGV that $K_{n,z}$ only has nonzero cohomology in degrees $[-m, m]\cup[-m-n, m-n]$. Since this holds for all closed points of $X\times Y$, we conclude K_n only has nonzero cohomology sheaves in degrees $[-m, m]\cup[-m-n, m-n]$. In exactly the same way we see that the maps $K_n \to K_{n+1}$ are isomorphisms on cohomology sheaves in degrees $[-m, m]$ for $n > 2m$.

If X and Y are smooth over k, then $X \times Y$ is smooth over k and hence regular by Varieties, Lemma 056S. Thus we will obtain the direct sum decomposition of K_n as soon as $n > 2m + \dim(X \times Y)$ from Lemma 0FZ4. The final statement is clear from this. □